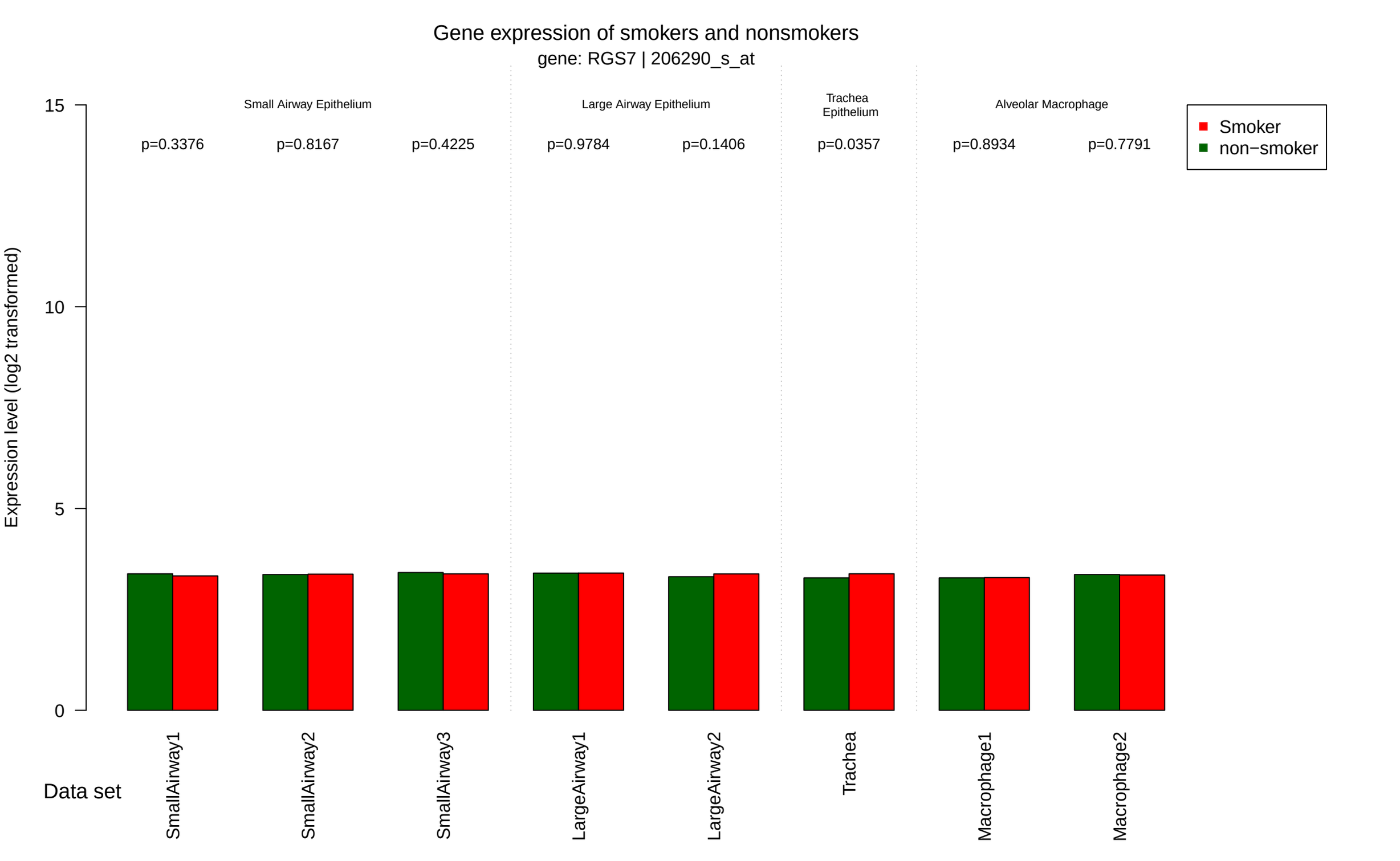
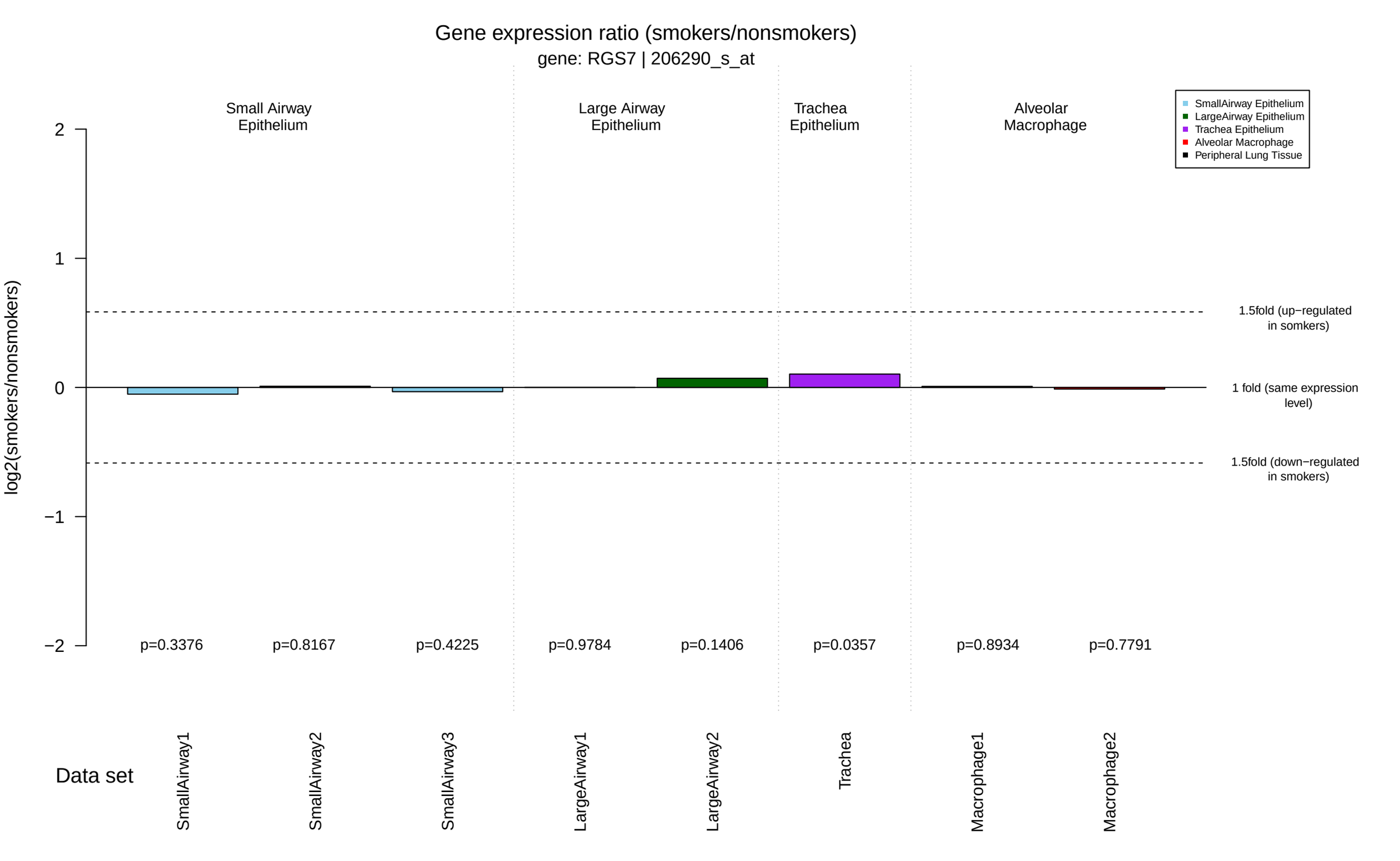
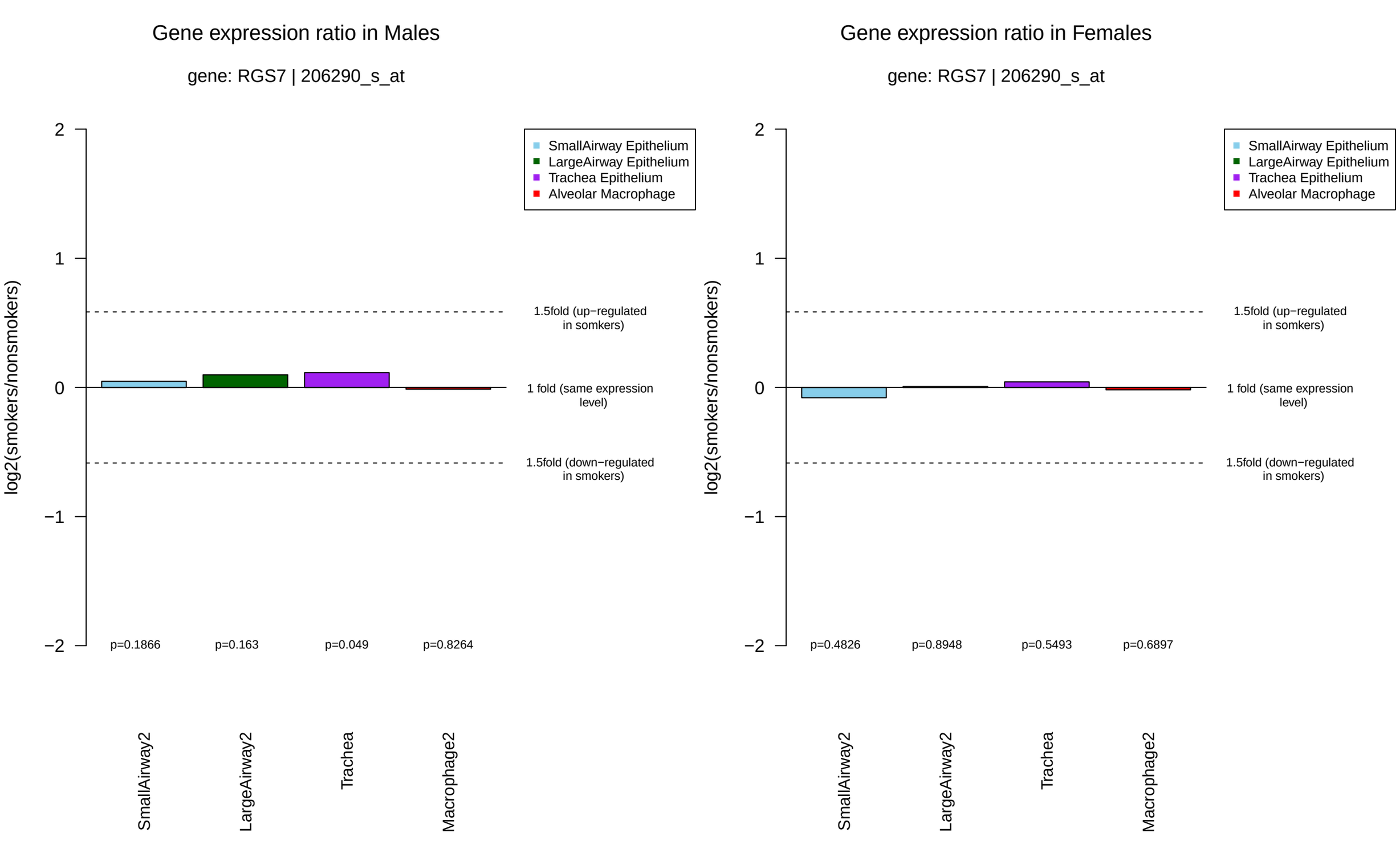
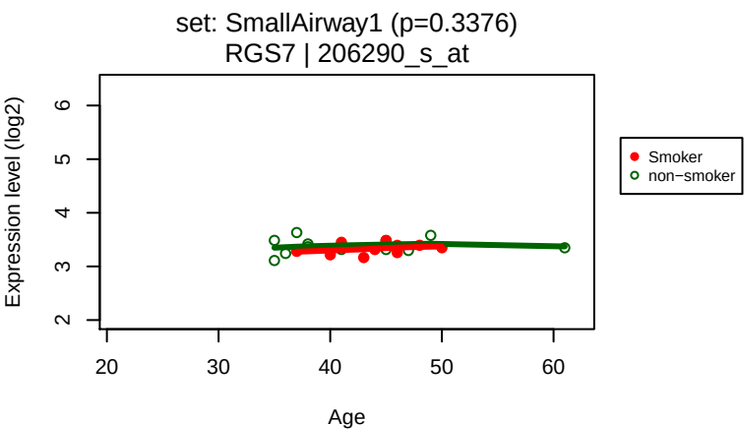
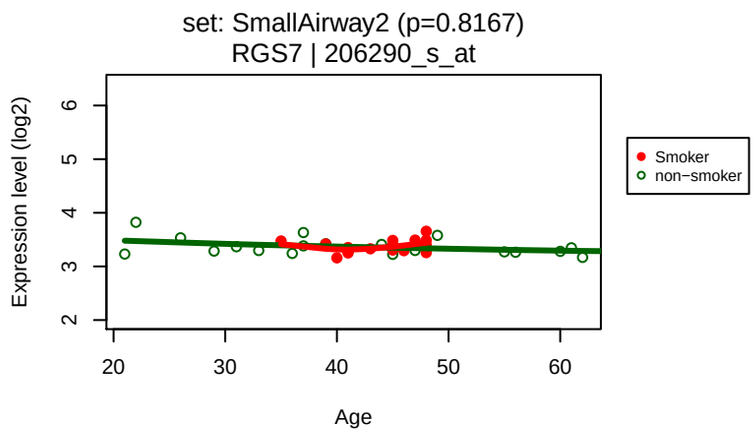
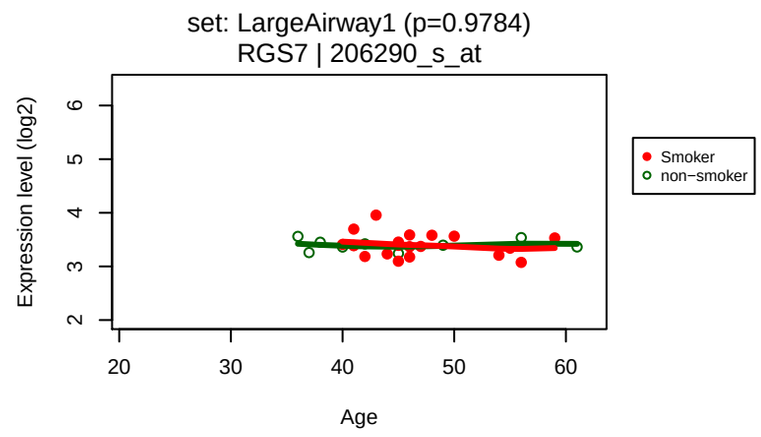

Gene expression of smokers and nonsmokers
gene: RGS7 | 206290_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3376
p=0.8167
p=0.4225
p=0.9784
p=0.1406
p=0.0357
p=0.8934
p=0.7791
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

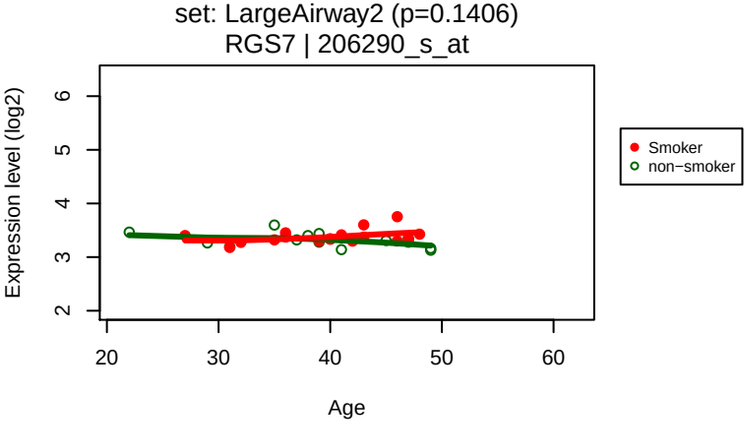

Gene expression ratio (smokers/nonsmokers)
gene: RGS7 | 206290_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3376
p=0.8167
p=0.4225
p=0.9784
p=0.1406
p=0.0357
p=0.8934
p=0.7791
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

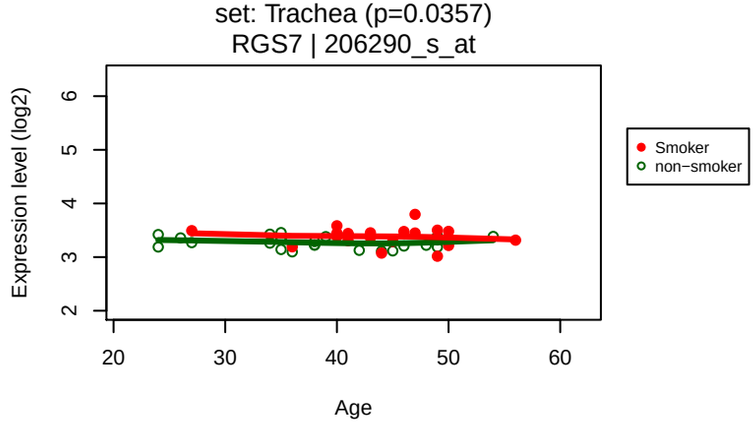

Gene expression ratio in Males
gene: RGS7 | 206290_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1866
p=0.163
p=0.049
p=0.8264
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RGS7 | 206290_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4826
p=0.8948
p=0.5493
p=0.6897
SmallAirway2
LargeAirway2
Trachea
Macrophage2

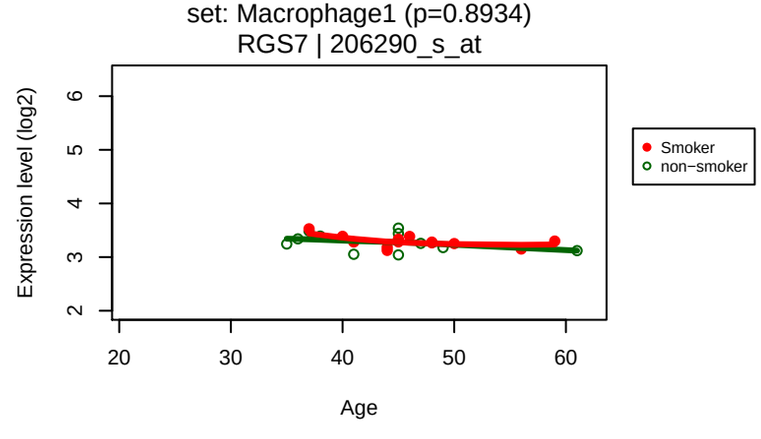

set: SmallAirway1 (p=0.3376)
RGS7 | 206290_s_at
Expression level (log2)
Age
Smoker
non−smoker

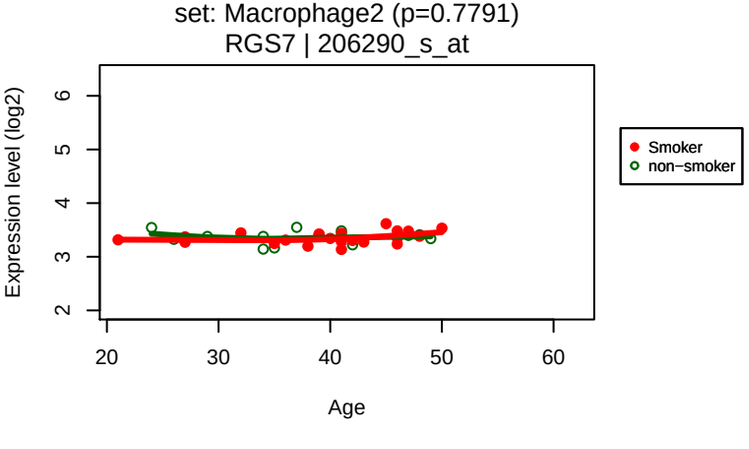

set: SmallAirway2 (p=0.8167)
RGS7 | 206290_s_at
Expression level (log2)
Age
Smoker
non−smoker

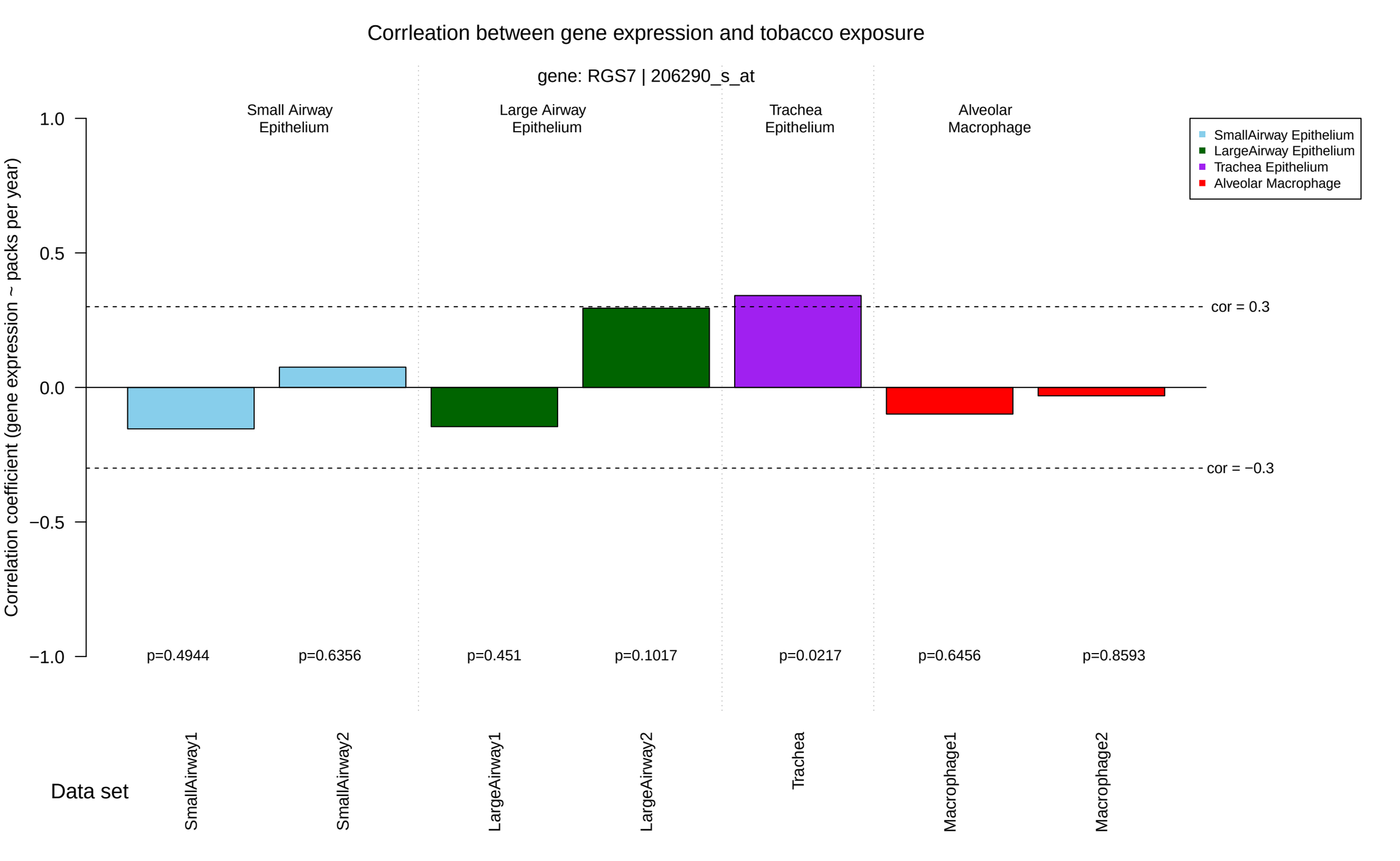

set: LargeAirway1 (p=0.9784)
RGS7 | 206290_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1406)
RGS7 | 206290_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0357)
RGS7 | 206290_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8934)
RGS7 | 206290_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7791)
RGS7 | 206290_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RGS7 | 206290_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4944
p=0.6356
p=0.451
p=0.1017
p=0.0217
p=0.6456
p=0.8593
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2